Investigating the Impact of Generative AI Tutoring Styles on Student Knowledge Acquisition and Confidence Calibration

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1 Introduction

The rise of Generative AI (GenAI) and its capabilities has become a major topic of discussion in the education industry [1]. Ever since the release of ChatGPT by OpenAI in November 2022, the education landscape has evolved tremendously [1]. Whereas institutions initially banned the use of AI, the trend has shifted to embracing and integrating GenAI into pedagogy [2]. This transition is supported by GenAI's ability to deliver personalized learning and customized feedback based on individual student needs [3]. By utilizing GenAI, students are now able to focus on higher-order skills such as critical thinking and creativity, while AI scaffolds and automates repetitive tasks [4, 5]. Moreover, due to today's rapidly evolving professional landscape, the adoption of GenAI in higher education has become a necessity rather than an option [6]. Hence, institutions need to create an environment where AI can be used to enhance human intelligence rather than replace it [1, 5].

However, the adoption of GenAI in higher education comes with a significant set of challenges. As previously mentioned, GenAI can be useful in taking over repetitive tasks [7], but this also leads to "cognitive offloading" where students outsource critical cognitive tasks to the AI [7]. The pattern of reliance starts from using AI as an information verification tool to slowly becoming a content delegation tool, causing students to blindly rely on AI to generate the final output [8]. It becomes even more concerning as AI creates a "productivity paradox", where despite students experiencing increased productivity, they lack depth in critical analysis [9]. For instance, due to convenience and speed that AI summaries provide, students tend to prioritize these summaries over verifying the underlying sources [10]. Despite the faster drafting process, the final result ended up being lower quality, with generic essays and a lack of critical analysis [1, 9]. Moreover, this matter is further supported by the so-called "death of essays", a phenomenon that has risen to prominence since COVID-19 due to the increase in take-home exams and students utilizing AI to cheat and plagiarize [1]. This raises the importance of AI literacy. Studies show that students with higher AI literacy tend to take a collaborative approach by planning and evaluating AI outputs, which consequently leads to better outcomes compared to students with lower AI literacy, who tend to over-rely on AI just to get the job done [11]. Additionally, over-reliance on AI can lead to a decline in face-to-face interactions, negatively impacting students' interpersonal skills and emotional intelligence. Thus, increasing the risk of social anxiety and isolation [12].

As previously mentioned, over-reliance on AI is a significant concern that must be addressed. AI tends to generate text with a convincing tone, giving an illusion that the output is always true, causing users to overlook hallucinations and in the end hallucinating themselves [1, 10]. To combat this, students are required to possess foundational knowledge before adopting the tool [1, 13]. This ensures they are able to critically validate the AI's information, as the human in the loop is responsible for the final output produced [1, 14]. Therefore, students must be taught to verify AI outputs [15]. On the other hand, educators are encouraged to create critical thinking assessments that can't be replicated by AI, such as video presentations or pitches [15, 1].

This gap creates a strong opportunity for designing how students should interact with AI. As previously mentioned, AI tutoring styles are not static [3]. The characteristics of the AI's interaction can be engineered using prompts (prompt engineering) and guardrails [16]. In other words, poor learning outcomes are not solely based on the AI's accuracy and performance, but also an issue with the interaction design itself [17, 18].

To address the illusion of truth, AI tools must be designed with guardrails that encourage cognitive engagement rather than handing out direct answers [19]. In the context of learning, the design of the AI interaction itself is more important than its capabilities [18]. Evidence shows that the use of scaffolding techniques, such as guiding hints and follow-up questions, are able to significantly improve students' learning outcomes [18]. AI chatbots with these guardrails are referred to as Socratic chatbots, which are designed to prevent giving out direct answers to encourage self-reflection and deep understanding of the topic [13, 20, 21]. When compared to default GenAI models, Socratic chatbots are able to produce 79% more learning improvements than the default models (127% vs. 48%) during practice assessment [18]. Additionally, unlike the default model where students experience learning loss when AI is removed, students did not experience any learning loss when the Socratic chatbot is removed [18]. This indicates that the default AI interaction may have short-term benefits, but confidence outpaces actual understanding, creating an illusion of competence [18]. Although the Socratic tutoring style works in many contexts, it is not universally superior [22]. In the context of coding, a study comparing the effects of Socratic AI versus normal AI shows that Socratic AI produces a higher error ratio and has no evidence of improving long-term outcomes [22]. Furthermore, learners with lower AI literacy tend to find scaffolding bots frustrating, which further amplifies the need for alternative tutoring styles [13]. This underscores the need to look beyond Socratic tutoring. It is important to explore other tutoring styles that could effectively improve both learning and confidence calibration.

Yet most literature rarely compares alternative interaction designs within the same study. Despite growing evidence on AI-supported learning, most studies evaluate one tutoring style at a time. As a result, the field still lacks clear comparative evidence on which tutoring style best results in balanced learning outcomes as well as psychological learning benefits. In practice, GenAI can be deployed with a spectrum of tutoring styles using prompt engineering

and guardrails [16]. Beyond Socratic approaches, evidence shows that reverse tutoring can be effective in boosting cognitive engagement and knowledge mastery [23, 24, 17, 25]. This pedagogical approach embodies the theory of learning-by-teaching by using recursive feedback that observes how a tutee applies what has been taught [23]. This approach is similar to designs where the AI takes the role of a student, prompting the human learner to teach the AI, thereby reinforcing the learner's own understanding [15]. Studies show that reverse tutoring creates the protégé effect where students learn harder when they know they must teach [23, 25]. There are two mechanisms that align with reverse tutoring [24]:

1. Generative processing: Students are encouraged to inherently organize and integrate their knowledge.
2. Metacognitive monitoring: Students are able to detect explanation gaps, which trigger self-correction.

Moreover, this socio-psychological interaction shifts the learners mindset to dig deeper into the topic, creating a more flexible mastery of the topic [24, 17].

Hence, in addition to the Socratic tutoring style, this study introduces the reverse tutoring technique. This study compares the three tutoring styles (direct-answer, Socratic, and reverse tutoring) within a single comparative framework, while assessing both cognitive and psychological outcomes. In particular, reverse tutoring, which is grounded in learning-by-teaching theory, despite showing clear effectiveness in improving learning outcomes, remains under-explored in the GenAI context. Therefore, this study aims to fill that gap by asking how different tutoring styles compare in their effects on learning performance, knowledge acquisition, self-efficacy, and confidence calibration. This combined evaluation is important as prior work shows that students can appear more productive with AI but could lose their learning once the tool is removed [9, 18]. Accordingly, this study aims to determine which tutoring style best supports durable learning rather than short-term productivity.

2 Literature Review

2.1 Generative AI in Education

It cannot be denied that AI has revolutionised the way the education industry operates [6]. In fact, AI has been implemented within the education industry for decades [21]. Introduced by Jaime R. Carbonell in 1970, the SCHOLAR tutor system was the first intelligent tutoring system (ITS) [26]. By utilizing a semantic network of concepts and facts, the system was able to evaluate students' knowledge of geography, creating a two-way interaction with the student [26]. Over the past decade, specialized AI tools such as Wolfram Alpha and Geogebra have enhanced STEM learning by increasing interactivity, while tools like Grammarly have assisted students in writing [21]. Although these tools were effective, they lack the conversational capabilities that current GenAI models possess. Driven by natural language processing (NLP), these large language models (LLMs) are able to hold sustained dialogue, and have since changed the way humans interact with information [21, 1].

The presence of GenAI has not only changed the way students and teachers interact, but also the entire curriculum at all levels of education [15, 2, 27]. From initially banning the use of AI due to concerns of cheating and plagiarism, institutions are now embracing and integrating AI into their pedagogy [2, 6]. This shift is supported by GenAI's ability to deliver an adaptive and personalized learning experience for students [3]. For a long time, education has been challenged in providing the unique needs of each individual student [21]. Evidence shows that students experience an increase in their learning outcomes when being tutored in a one-on-one setting [28]. However, the cost of providing human one-on-one tutoring is expensive in both financial and human resources, especially as teachers are also burdened with administrative responsibilities [28, 15]. This is where GenAI comes into play by providing a scalable and cost-effective one-on-one tutor compared to traditional human tutors that cost $3,500 - -4$,300 per student per year [28]. Hence, lowering the entry barrier for complex subjects by providing examples and analogies that cater to the user's understanding and interest [2, 29]. For example, in the context of vocabulary learning, the AI's output can be personalized based on the user's hobbies such as cooking, sci-fi, or sports; turning a dry learning experience into a more engaging and interactive one [29, 1]. Additionally, AI aids overworked instructors by making complex pedagogy accessible [15]. Instructors are able to leverage AI to provide examples and analogies to address misconceptions and design materials that reinforce learning [15].

Despite the immense benefits that GenAI provides, when used without proper guidance it can cause harm to long-term learning outcomes [18]. Therefore, it is critical for educators to standardize GenAI's use in education. In order to correctly adopt GenAI in their pedagogy, educators are encouraged to follow UNESCO's AI Competency Framework, which acts as a guide for AI use and misuse [30]. The framework highlights several key dimensions essential to effective AI implementation: students must develop foundational knowledge, use the tool ethically, and ensure that a human remains responsible for overseeing and validating the final output [30, 14]. Thus, institutions must create an environment where AI enhances human intelligence rather than replacing it, fostering higher-order thinking and

creativity [5, 1]. Moreover, GenAI should be prevented if it removes the opportunity for students to develop cognitive abilities or social skills [1].

However, in practice, GenAI is still being used incorrectly, leading to short-term productivity but long-term learning loss, known as the "productivity paradox" [9, 18]. This phenomenon occurs because students tend to offload cognitive tasks to the AI, such as planning, organising, and evaluating [7]. Evidence shows that most students prioritize the convenience and speed of AI summaries over verifying the underlying sources [10]. Hence, students who over-rely on AI are found to produce generic and lower-quality work [1]. Consequently, the excessive use of ChatGPT reduces their academic performance and memory [31]. This shows that GenAI cannot yet be supportive of self-regulated learning unless it is pedagogically engineered to provide hints and scaffold rather than direct answers [7]. To mitigate the misuse of GenAI, educators are encouraged to provide assessments that require students to document their process in reaching their final output (e.g. oral examination or AI-output documentation)[27, 1]. Furthermore, institutions are encouraged to use GenAI to foster higher-order skills and creativity, combining traditional pedagogy with AI-assisted exploration [5, 2]. Thus, it is important to design a human-centric model where GenAI acts as a cognitive partner rather than a replacement [5].

2.1.1 Socratic AI versus Direct AI

As previously mentioned, without proper guardrails, GenAI outputs direct answers, imposing the risk of cognitive offloading [7, 5]. To prevent this, Socratic chatbots are designed by using prompt engineering and guardrails to encourage cognitive engagement by offering hints and guidance rather than direct answers [16, 20]. As the name suggests, the design of Socratic AI chatbots originated from the Socratic teaching method [21]. The method itself was originally designed to doubt all assumptions using questions and dialogues, forcing students to construct their own knowledge [32]. This technique proves to be effective in improving students' learning outcomes, improving their learning gain by 59% in the context of coding comprehension [33]. In addition, a comparative study comparing Socratic AI versus direct AI shows that Socratic AI is able to improve learning gain by 127% with no learning loss when the AI is removed [18]. In comparison, the default model shows a learning loss of 17% when the AI is removed [18]. The open-ended questions and reflective prompts are proven to be more useful in promoting self-regulated learning as agreed by 85% of educators [34]. This is due to the back-and-forth interaction that encourages students to verbalise their reasoning and correct any misconceptions in comparison to the default model that gives direct solutions that deprive students of cognitive effort [35]. Thus, it can be acknowledged that the design of the AI interaction is more important than the model capabilities in improving learning outcomes [18].

Additionally, the convincing tone of default AI models can lead to users believing every response is true, despite the inaccuracies caused by hallucinations [1, 36]. This "illusion of truth" can be harmful, as it leads to users overlooking these hallucinations, causing errors in their final output [1, 10]. Subsequently, users' confidence tends to align with the AI-expressed confidence, further imposing the danger of this illusion [37]. Moreover, the authoritative tone of the AI leads to students being over-reliant on the AI, which is proven to limit creativity and critical thinking as well as social and emotional development [11, 37]. Furthermore, being over-reliant on the AI leads to a lack of face-to-face interaction which negatively impacts students' interpersonal skills that leads to a higher risk of social anxiety and isolation [12].

To mitigate this, Lo et al. proposed a CLEAR framework, which stands for Critical, Reflective, Adaptive, Explainable, and Responsible [14]. This framework encourages students to critically analyse the AI's output, reflect on their own learning, adapt to the feedback, explain the reasoning behind their answer, and be responsible for the final output [14]. However, to be able to apply this framework, the student must possess foundational knowledge of the topic as well as AI literacy to critically evaluate the AI's output [14]. Simply flagging whether the content is fake is not effective as students must develop internal critical thinking to succeed [38]. A study comparing traditional study methods, studying with direct AI, and studying with Socratic AI approaches, found that users who study with direct AI showed a learning loss of 17% when the AI is removed, while users who study with the traditional method and Socratic AI showed no learning loss [18]. This further proves that Socratic AI chatbots go beyond fact-checking, as it allows students to recognize cognitive biases and build emotional triggers, encouraging them to become more cognitively active [38].

Despite the effectiveness in Socratic AI chatbots, it is not a one-size-fits-all solution. Learners with low AI literacy tend to find these scaffolding bots frustrating [13]. When compared to direct AI in the context of coding, students who were assigned the Socratic AI actually performed worse with lower efficiency and higher error ratio [22]. Additionally, no significant long-term learning outcome was found, meaning the domain and task matter [22]. Hence, in order to achieve productive human-AI collaboration, students must possess a high AI literacy as a prerequisite [13, 11]. Studies show that students with higher AI literacy are able to evaluate, and steer the AI to receive deeper insights, while lower literacy learners revert to simple queries [13, 11]. Therefore, it is necessary for educators to provide prompt literacy to students, fostering UNESCO's AI Competency Framework that envisions students as AI co-creators [5, 30]. As previously

acknowledged, each student has their own unique needs [21]. Therefore, research on designing other human-centric models should also be pursued to cater to different learning styles and needs [5].

2.2 Various AI Learning Strategies

Beyond Socratic AI chatbot, students are able to engage with AI through a variety of interaction styles, each offering their own benefits and risks as shown in Table 1 [15].

AI Style	Description	Benefit	Risk
Mentor	Provides specific feedback and guidance on the student's work	The frequent feedback allows student to reflect even if not all advice are taken	May encourage over-reliance where students do not critically evaluate the AI's feedback
Tutor	Offers direct instruction and answers	Effective for quick acquisition of knowledge	The uneven knowledge base could cause hallucinations
Coach	Engages with students metacognitively (think about their own thinking) by prompting reflections	Builds self-regulation and planning skills	Incorrect advice may lead to misguidance; the interaction style may not suit every student
Teammate	Collaborates with student to solve tasks by playing a devil's advocate role to challenge the student's assumptions	Provides alternate perspectives and fosters collaborative learning	May provide errors that misguide the student's understanding
Student	Reverses the role, the student teaches the AI	Promotes learning-by-teaching which can uncover knowledge gaps	AI's argumentative behavior may derail the benefits of this approach
Simulator	Provides realistic role-play scenarios to practice the student's skills	Practical and hands-on knowledge application aids in transfer of learning	The story element may distract from the core concept being taught
Tool	AI to extend the learner's capabilities by being a general-purpose utility	More work can be done within the same timeframe	Risk of outsourcing the thinking process to the AI

Table 1: Various Learning Approaches with AI [15].

Additionally, the ICAP framework [39] gives a clear categorization on different levels of cognitive engagement:

1. Interactive: Learning by collaboration where both parties contribute by building on each other's ideas (e.g. debate).
2. Constructive: Generative new information beyond the materials provided (e.g. mind mapping).
3. Active: Learning using physical manipulation without generating anything new (e.g. highlighting).
4. Passive: Receiving information without any physical engagement (e.g. listening to a lecture).

The ICAP framework has shown that as students move higher in the hierarchy from passive to interactive, learners will gain an increase in learning due to the knowledge-change processes [39]. Drawing upon the different ways students are able to engage with AI and its benefits, as well as the validity of the ICAP framework, this study proposes to compare three different AI tutoring interactions: direct-answer, Socratic, and reverse-tutoring. The reverse-tutoring approach is chosen due to its potential in fostering active learning which has been proven to increase learning outcomes [39].

2.2.1 Reverse-tutoring (The Protégé Effect)

"While we teach, we learn" is a famous quote by the Roman philosopher Seneca [25]. Beyond the quote, decades of research have shown that learning-by-teaching has proven effective in improving one's knowledge on a topic [25, 40].

The beneficial learning outcomes of learning-by-teaching are driven by socio-psychological mechanisms whereby being assigned a teacher role increases their agency, consequently boosting their cognitive engagement and outcome [25, 23]. This phenomenon is known as the "protégé effect" where students tend to learn harder as they know they must teach [23]. The phenomenon is further supported by Fiorella and Logan's study that found that the process of reverse-tutoring underwent two mechanisms: generative processing and metacognitive monitoring [24].

- **Generative processing:** Learners generate new information beyond the materials provided and draw relationships between them. This process is driven by the learner's own understanding and knowledge.
- **Metacognitive monitoring:** As the learners teach the material they constantly monitor their own understanding, detecting explanation gaps.

The combination of these two mechanisms shifts learners' knowledge from surface-level understanding to a flexible mastery of the material [24]. These processes are aligned with the interactive level of engagement from the ICAP framework, as students are able to reflect on their own understanding and build upon knowledge gaps as they teach [39, 24]. Additionally, a comparative study by Kobayashi found that students who were explicitly told they were going to teach performed twice as well as those who were not told [40]. Moreover, students who later actually taught, performed even better than those on preparation-only conditions, indicating that the act of communicating knowledge possesses a unique cognitive benefit beyond preparation alone [40].

One of the most frequently cited teachable agent (TA) is Betty's Brain (2007), a pedagogical agent built to foster reasoning for middle school students in the subject of biology [41]. Dialogue-based interaction was not possible in this era due to the lack of advancement in NLP [41]. Instead, students interact by building causal maps that act as the "brain" of the agent [41]. The study shows that the presence of the TA was able to absorb the blame of errors, allowing students to acknowledge their mistakes while protecting their ego [41]. TAs are able to act as an ego-protective buffer, where students often avoid critical feedback because it feels like a punishment, the presence of the TA allows students to receive feedback without feeling a sense of failure [42, 41]. Additionally, the TA's presence was incredibly beneficial for low-achieving students, where they were able to score equal to the high-achieving students on the hardest questions [42, 41]. Moreover, the benefits of TA persisted even after the technology was removed, drawing similarities to Socratic AI where learning outcome is retained even after the AI is removed [43, 18].

Putting the learning-by-teaching theory into the context of GenAI, we reverse the role of the student and the AI by having the student teach the AI as a naive tutee [15]. This can be done by designing the personality of the chatbot to be curious and open-minded [17]. By doing so, students are driven to give higher-order explanations and engage in more interactive exchanges [17]. The AI should act as a recursive feedback tutor that observes what the student has taught them [23]. A study by Jin et al., developed a TA of their own called AlgoBo, which acts as a struggling novice in the subject of coding [44]. AlgoBo is prompt engineered to have a strict knowledge of only documents within the chat, and acts as either a help-receiver or an active "questioner" that asks why and how questions to the student [44]. This strategy encourages students to think of edge-cases and misconceptions, just as denoted by the protégé effect [24, 44]. Moreover, despite leading students to produce deeper conversations, there was no increase in self-reported cognitive load [44].

It is important to note that the personality of these AIs matters [17]. If the AI fails to emphasise a curiosity by showing agreeable and extroverted behavior, it may lead to shallow and repetitive exchanges [17]. Even though it may lead to an increase in interaction volume, the interaction is not driven by higher-order engagement [17]. Therefore, reverse-tutoring is not a one-size-fits-all solution and it depends on the TA's persona design [17].

2.3 Confidence Calibration

Confidence calibration is defined as how well a person's subjective sense of certainty matches actual correctness [45]. In the context of human-AI interaction, confidence calibration is usually refers to how much a user trusts the AI's output in comparison to how accurate the output is [45]. As previously mentioned, the illusion of truth, where modern LLMs can act highly confident even when they output incorrect information, poses a huge risk of causing miscalibration [46, 1]. A study by Zhang et al. shows that users tend to rely more on the AI when the AI acts more confidently, causing users to overlook the AI's hallucinations [47]. This is especially dangerous for users with low AI literacy, as they might take in information without critically evaluating it, and eventually become over-reliant on the AI [11]. Furthermore, when miscalibration occurs, users tend to blame themselves for the mistake, destroying their self-efficacy, and eventually lean harder on the AI [45].

A consistent pattern across these studies is that calibration improves when the system shows uncertainty (e.g. asking the user to double-check), and when the human themselves can complement the AI's output [48, 46]. This uncertainty friction leads users to fact-check with their own knowledge or external sources, leading to a deeper and accurate final output [46]. Thus, designing an AI system that fosters uncertainty and prompting the user to fact-check with their own

knowledge is crucial in achieving a better calibration [46]. Additionally, literacy plays a crucial role in the interaction, as users with low literacy may find the uncertainty friction overwhelming [13]. Hence, AI usage education such as the CLEAR framework [14] is crucial in making sure users are able to understand the uncertainty and fact-check with their own knowledge.

In the scope of this study, the different designs of AI tutoring styles will be compared in terms of their calibration performance, to determine which tutoring style is most effective in fostering a better calibration. The default direct-answer AI risks strong confidence inflation due to the lack of uncertainty friction, leading to over-reliance on the AI's output [18]. Socratic AI may improve calibration by encouraging self-explanation and error checking, but may raise frustration [13, 18]. Reverse-tutoring can surface knowledge gaps and reduce the illusion of competence by metacognitive monitoring [24]. Finally, the interaction between calibration and self-efficacy is important to take note of, as it is proven to be a vicious cycle where bad AI advice can reduce the user's self-efficacy, which can then increase future dependence on the AI [45].

In measuring calibration performance, it is important to measure the alignment between students' level of confidence and their actual learning results, especially when the AI is removed. Following prior work, the use of multiple assessments (pre-test, AI-assisted practice, and post-test without AI) can be used to distinguish short-term task completion from retained knowledge and well-calibrated confidence [18].

3 Research Question

The main research question in this study is: How do (A) tutoring style factors influence the effectiveness of generative AI tutoring; specifically, how does it affect (C) student learning performance, (D) knowledge acquisition, (E) self-efficacy, and (F) confidence calibration under conditions of varying perceived task difficulty?

The tutoring style factors examined in this research are *(A1) direct-answer AI tutoring, (A2) Socratic AI tutoring, and (A3) reverse-tutoring (protégé effect, where the student teaches the AI acting as a naive tutee).* The outcomes measured in the study are *(C1) learning performance (objective test scores), (D1) knowledge acquisition (knowledge gain from pre- to post-intervention), (E1) self-efficacy (confidence in one's ability), and (F1) confidence calibration (alignment between confidence and actual correctness).*

This study hypothesizes that reverse-tutoring will lead to better knowledge mastery, stronger self-efficacy, and better alignment between confidence and actual competence than both the Socratic chatbot and the direct-answer chatbot, as teaching forces learners to organise knowledge and expose gaps, reducing the illusion of competence while supporting deeper learning.

4 Methodology

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